AUDIOSHIELD FORENSICS

Audio Forensic Analysis Report

Report ID	ASF-20260916-C399
Generated	2026-09-16 23:25:11 India Standard Time
File analyzed	884d061a-8db5-473b-9e85-a89fd71b2682.mp3

This report presents separate evidence streams and conservative forensic indicators. It is not a substitute for expert examination of the original evidence.

1. Executive Summary

Overall risk assessment: **medium**

Overall assessment: **inconclusive** | Confidence: **0.9825613765869256**

Key findings

- AI consensus: spoof (risk: high).
- Metadata risk: low; 1 finding(s).
- Signal risk: low; 0 anomaly/anomalies.

Official Eval AI comparison: CNN 69.83%, SVM 89.40%, AASIST 86.45%.

Recommendation: Evidence is mixed or incomplete; requires further investigation.

2. File Information

Property	Value
File name	884d061a-8db5-473b-9e85-a89fd71b2682.mp3
File size (bytes)	103070
Format	mp3
Duration (seconds)	6.391293
Sample rate (Hz)	44100
Channels	2
Codec	mp3
Bitrate (bits/sec)	129013
Creation time	2026:09:16 23:25:05+05:30
Modification time	2026:09:16 23:25:05+05:30

3. AI Detection Results

Official Eval-set comparison

Model	Accuracy	F1	EER	Rank
CNN	69.83%	0.7978	0.0716	3rd
SVM	89.40%	0.9400	0.1698	1st
AASIST	86.45%	0.9184	0.0720	2nd

Best model: **SVM** (best balanced rank across accuracy, F1, and EER; lower EER is better).

4. Metadata Forensics

Extracted metadata

Field	Value
File Name	884d061a-8db5-473b-9e85-a89fd71b2682.mp3

Field	Value
File Size	103070
Format	mp3
Duration	6.391293
Codec	mp3
Sample Rate	44100
Channels	2
Bitrate	129013
Encoder	Lavf57.83.100
Creation Time	2026:09:16 23:25:05+05:30
Modification Time	2026:09:16 23:25:05+05:30
Device Info	Not available
Software	Lavf57.83.100
Comments	Not available

Risk score: **low**

Findings

Severity	Field	Description
low	device	Device information is missing; provenance cannot be verified from metadata alone.

Metadata indicators are not conclusive on their own and require further investigation.

5. Signal Forensics

Spectral features

Feature	Mean	Std	Min	Max
Spectral Centroid	1714.9495051387166	339.53180723169515	1109.788184629998	2331.8430451591885
Spectral Bandwidth	1882.2428895046417	167.99507411765657	1490.601348155279	2307.922093088411
Spectral Rolloff	3660.1313694267515	558.547814191135	2359.375	5265.625
Spectral Flatness	0.009546121773695584	0.010801809044550406	0.000630763650406152	0.068091697990894432
Spectral Contrast	25.83047356001864	5.005134634916103	17.908159819806777	37.52062725526791

Temporal features

Feature	Mean	Std	Min	Max
Zero Crossing Rate	0.11831334593949044	0.04958857670177789	0.03857421875	0.21337890625
Rms Energy	0.07833978899155453	0.04176374168504122	0.02412654086947441	0.17328576743602753
Pitch	76.32756385103076	23.046867607929887	65.40639132514966	130.8127826502993

Risk score: **low**

Anomalies

No signal-level anomalies were reported.

6. Evidence Fusion

Evidence source	Risk / result
AI	spoof (risk: high)
Metadata	low
Signal	low
Overall	medium

Assessment: **inconclusive** | Confidence: **0.9825613765869256**

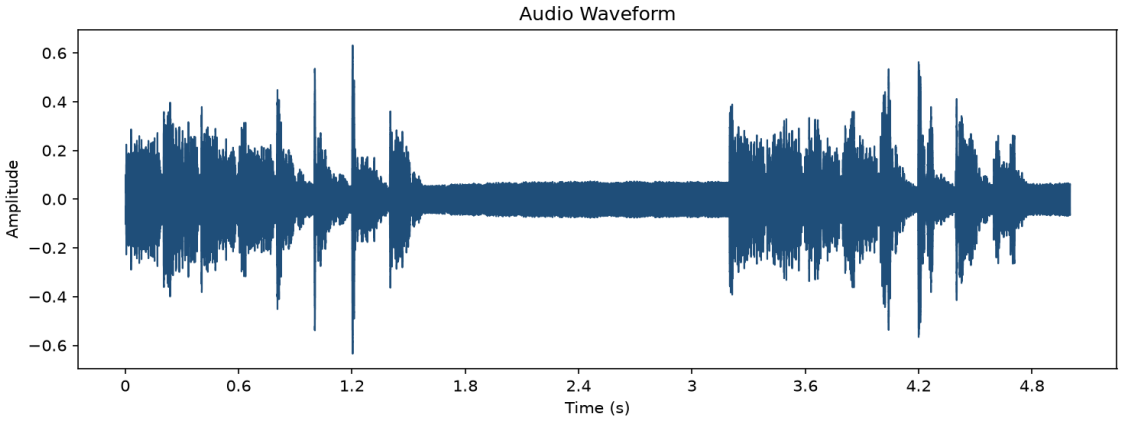
Reasoning: Source risks are AI=high, metadata=low, signal=low; applied documented rule-based logic.

Recommendation: Evidence is mixed or incomplete; requires further investigation.

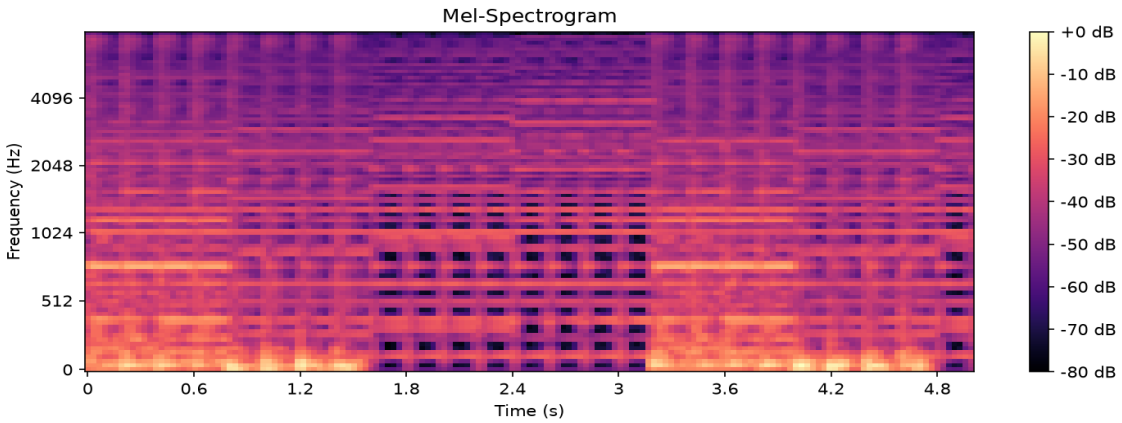
AI model breakdown

Model	Label	Confidence
CNN	spoof	1.0
SVM	spoof	0.9962259736519202
AASIST	spoof	0.9514581561088562

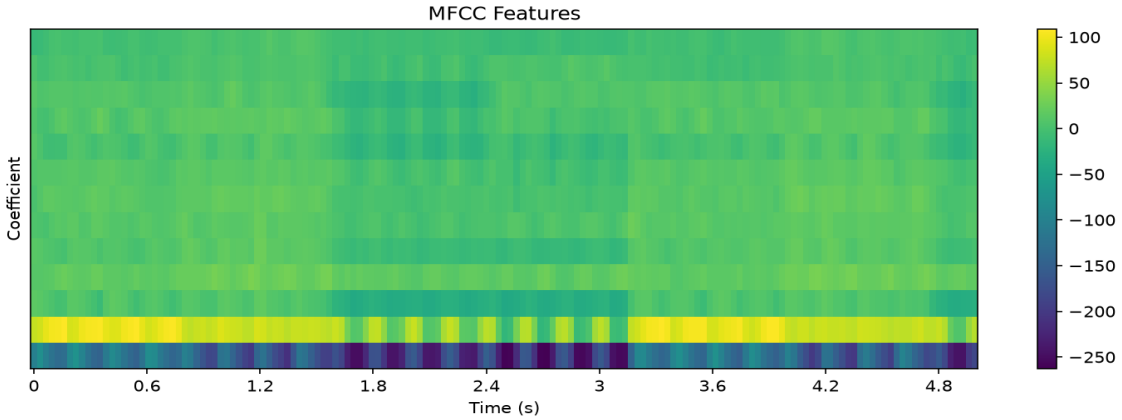
7. Visual Analysis



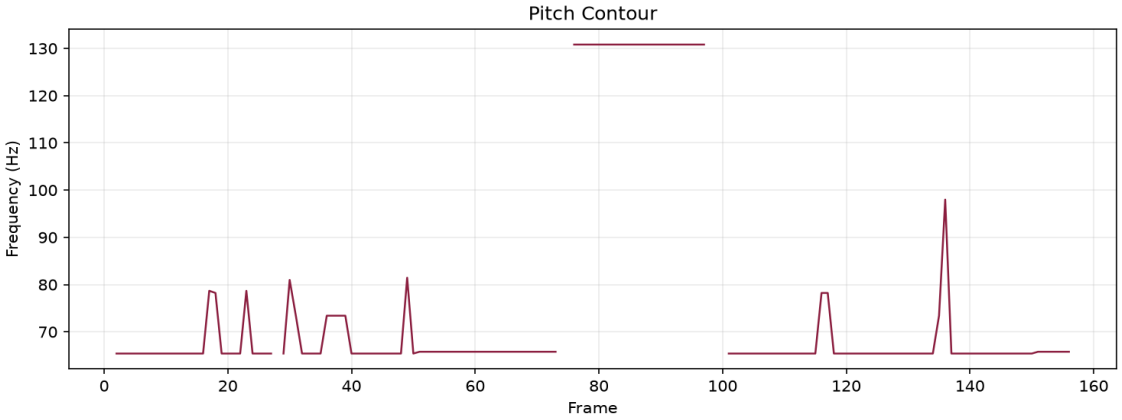
Waveform: Time-domain waveform.



Spectrogram: Frequency content over time.



MFCC: Mel-frequency cepstral coefficients.



Pitch contour: Estimated fundamental frequency.

8. Conclusion

The combined evidence indicates an overall assessment of **inconclusive** with a reported risk level of **medium**. These are indicators from the available reports and should be interpreted with the original audio and chain of custody.

Limitations: AI results may be unavailable; metadata can be edited or lost; signal features are influenced by recording conditions and preprocessing; and this report does not establish authenticity by itself.

9. Appendix

Raw AI evaluation results

{ "CNN": { "model": "CNN", "checkpoint":
"C:\\Users\\allen\\AudioShield-Forensics\\models\\cnn\\cnn_final_best.pth", "dataset": "ASVspoof 2019 LA",
"split": "eval", "total_samples": 71237, "bonafide_samples": 7355, "spoof_samples": 63882, "best_epoch": 4,
"best_validation_loss": 0.003000220499216719, "threshold": { "value": 0.40700000000000003, "source":
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"selection_source": "dev_only", "path":
"C:\\Users\\allen\\AudioShield-Forensics\\results\\cnn_final_threshold.json" }, "normalization": { "method":
"min-max", "train_min": -80.0, "train_max": 0.0, "source":
"C:\\Users\\allen\\AudioShield-Forensics\\results\\cnn_final_norm.json" }, "threshold_independent_metrics":
{ "roc_auc": 0.9800478133427984, "eer": 0.0716185237520802, "eer_percent": 7.161852375208019 }, "metrics": {
"accuracy": 0.6982888105900024, "precision": 0.9998820726903936, "recall": 0.6636298174759713, "f1_score":
0.7977719441856964, "specificity": 0.9993201903467029, "fpr": 0.0006798096532970768, "tpr":
0.6636298174759713, "balanced_accuracy": 0.8314750039113371, "threshold": 0.40700000000000003,
"selection_method": "balanced_accuracy_with_spoof_recall_constraint" }, "confusion_matrix": { "labels": [
"bonafide", "spoof"], "matrix": [[7350, 5], [21488, 42394]] } }, "SVM": { "model": "SVM", "split":
"official_eval", "total_samples": 71237, "bonafide_samples": 7355, "spoof_samples": 63882, "metrics": {
"accuracy": 0.8940017125931748, "precision": 0.9543628708319218, "recall": 0.926082464544003, "f1_score":
0.9400100102485879, "roc_auc": 0.874909277304299, "eer": 0.16983042600362058 } }, "AASIST": {
"total_samples": 71237, "bonafide_samples": 7355, "spoof_samples": 63882, "tn": 7242, "fp": 113, "fn": 9540,
"tp": 54342, "accuracy": 0.8644945744486713, "precision": 0.9979248921127537, "recall": 0.850662158354466,
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} }

Raw metadata report

{ "metadata": { "file_name": "884d061a-8db5-473b-9e85-a89fd71b2682.mp3", "file_size": "103070", "format":
"mp3", "duration": "6.391293", "codec": "mp3", "sample_rate": "44100", "channels": 2, "bitrate": "129013",
"encoder": "Lavf57.83.100", "creation_time": "2026:09:16 23:25:05+05:30", "modification_time": "2026:09:16
23:25:05+05:30", "device_info": null, "software": "Lavf57.83.100", "comments": null, "raw_ffprobe": {
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"audio", "codec_tag_string": "[0][0][0][0]", "codec_tag": "0x0000", "mime_codec_string": "mp4a.40.34",
"sample_fmt": "fltp", "sample_rate": "44100", "channels": 2, "channel_layout": "stereo", "bits_per_sample":
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"start_pts": 353600, "start_time": "0.025057", "duration_ts": 90193920, "duration": "6.391293", "bit_rate":
"128000", "disposition": { "default": 0, "dub": 0, "original": 0, "comment": 0, "lyrics": 0, "karaoke": 0,
"forced": 0, "hearing_impaired": 0, "visual_impaired": 0, "clean_effects": 0, "attached_pic": 0,
"timed_thumbnails": 0, "non_diegetic": 0, "captions": 0, "descriptions": 0, "metadata": 0, "dependent": 0,
"still_image": 0, "multilayer": 0 }, "tags": { "encoder": "Lavc57.10" } }], "format": { "filename":
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1, "nb_programs": 0, "nb_stream_groups": 0, "format_name": "mp3", "format_long_name": "MP2/3 (MPEG audio
layer 2/3)", "start_time": "0.025057", "duration": "6.391293", "size": "103070", "bit_rate": "129013",
"probe_score": 52, "tags": { "encoder": "Lavf57.83.100" } } }, "raw_exiftool": { "SourceFile":
"C:/Users/allen/AudioShield-Forensics/uploads/884d061a-8db5-473b-9e85-a89fd71b2682.mp3", "ExifToolVersion":
13.59, "FileName": "884d061a-8db5-473b-9e85-a89fd71b2682.mp3", "Directory":
"C:/Users/allen/AudioShield-Forensics/uploads", "FileSize": "103 kB", "FileModifyDate": "2026:09:16
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23:25:05+05:30", "FilePermissions": "-rw-rw-rw-", "FileType": "MP3", "FileTypeExtension": "mp3", "MIMEType":
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"ChannelMode": "Stereo", "MSStereo": "Off", "IntensityStereo": "Off", "CopyrightFlag": false,
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"bitrate": "129013", "encoder": "Lavf57.83.100", "creation_time": "2026:09:16 23:25:05+05:30",
"modification_time": "2026:09:16 23:25:05+05:30", "device_info": null, "software": "Lavf57.83.100",
"comments": null }, "analysis": { "findings": [{ "field": "device", "has_issue": true, "severity": "low",
"description": "Device information is missing; provenance cannot be verified from metadata alone." }],
"risk_score": "low", "suspicious_fields": [], "missing_fields": ["device_info", "comments"] } }

Raw signal report

{ "spectral_features": { "spectral_centroid": { "mean": 1714.9495051387166, "std": 339.53180723169515, "min":
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167.99507411765657, "min": 1490.601348155279, "max": 2307.922093088411 }, "spectral_rolloff": { "mean":
3660.1313694267515, "std": 558.547814191135, "min": 2359.375, "max": 5265.625 }, "spectral_flatness": {
"mean": 0.009546121773695584, "std": 0.010801809044550406, "min": 0.000630763650406152, "max":
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17.908159819806777, "max": 37.52062725526791 } }, "temporal_features": { "zero_crossing_rate": { "mean":
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"mean": 0.07833978899155453, "std": 0.04176374168504122, "min": 0.02412654086947441, "max":
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65.40639132514966, "max": 130.8127826502993 } }, "anomalies": { "anomalies": [], "risk_score": "low",
"summary": "No broad signal-level anomalies identified." }, "temporal_analysis": { "consistency_score": 1.0,
"anomalies": [], "summary": "Temporal patterns are broadly consistent.", "pitch_analysis": {
"stability_score": 1.0, "anomalies": [], "description": "Pitch contour is broadly continuous." },
"energy_analysis": { "energy_mean": 0.07833979278802872, "energy_std": 0.04176373779773712, "anomalies": [] }
}, "plots": ["results\\signal_plots\\waveform.png", "results\\signal_plots\\spectrogram.png",
"results\\signal_plots\\mfcc.png", "results\\signal_plots\\pitch_contour.png"], "signal_risk": "low",
"summary": "Signal risk is low. No broad signal-level anomalies identified." }

Methodology

This report compiles the supplied AI, metadata, signal, and fusion JSON outputs. No new model inference is performed during report generation. Evidence sources remain separated to preserve interpretability.